STALKS OF F_! FOR LOCALLY QUASI-FINITE MORPHISMS AND EXACTNESS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-etale.tex; no statement or proof step was rewritten, reordered or removed. The counted span is the author environment and its own proof environment. References inside the span are handed to the assembler as navigation stubs that carry the canonical tag of the cited tag only; whether each of them is load-bearing is recorded on the stub itself.

Editorial note. Editorial note: the functoriality of the stalk formula in F is taken over from the statement of lemma-finite-support-stalk, whose proof asserts rather than verifies the corresponding functoriality; the present proof itself omits nothing.

Allowed standard prerequisites (editorial).

- Stalks of f_! with finite support (Lemma lemma-finite-support-stalk, canonical tag 0F6P). cited at proof lines 1390-1391. The stalk formula is immediate from and in fact equivalent to this lemma.
- Exactness checked on stalks (Etale Cohomology, Theorem etale-cohomology-theorem-exactness-stalks, canonical tag 03PU). cited at proof lines 1392-1394. Exactness of f_! follows from the direct sum stalk formula via this theorem.

Source reference. Original lemma 0F6P: sources/stacks-chow/tags.txt, source lines 18979–18979. Cited at proof lines 1390-1391: the stalk formula is immediate from and in fact equivalent to this lemma.

Source reference. Original theorem 03PU: sources/stacks-chow/tags.txt, source lines 4871–4871. Cited at proof lines 1392-1394: exactness of f_! follows from the direct sum stalk formula via this theorem.

Lemma 0.2. *Let $f : X \to Y$ be a locally quasi-finite morphism of schemes. Then*

(1) *for $\mathcal{F}$ in $Ab(X_{étale})$ and a geometric point $\overline{y} : \mathrm{Spec}(k) \to Y$ we have*

$$(f_!\mathcal{F})_{\overline{y}} = \bigoplus\nolimits_{f(\overline{x})=\overline{y}} \mathcal{F}_{\overline{x}}$$

functorially in $\mathcal{F}$, and

(2) *the functor $f_! : Ab(X_{étale}) \to Ab(Y_{étale})$ is exact and commutes with direct sums.*

Proof. The formula for the stalks is immediate (and in fact equivalent) to Lemma 0F6P. The exactness of the functor follows immediately from this and the fact that exactness may be checked on stalks, see Étale Cohomology, Theorem 03PU. □

Date: stacks-project:more-etale.tex:local-snapshot-20261012; GNU FDL 1.2 or later.